Journal of Cluster Science (2025) 36:91
https://doi.org/10.1007/s10876-025-02811-4

ORIGINAL PAPER

Pyridine and Cyclohexylamine Fluorescence Paper Sensors Based on Two cubane-type Cu_4I_4 Clusters

Xin-Yang Du[1] · Li Song[2] · Xin-Yu Wei[1] · Ying-Ying Zhang[1] · Yu-Xin Zhang[1] · Xia-Qun Zeng[1] · Hao-Nan Hu[1] · Zhou-Hang Jiang[1] · Yong Zhang[1] · Wen-Xiang Chai[1]

Received: 11 January 2025 / Accepted: 21 March 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract
A cubane-type Cu_4I_4 cluster 3-$PhPy_4Cu_4I_4$ (**1**) was designed and synthesized (3-PhPy = 3-phenylpyridine). The cluster was characterized by elemental analysis, X-ray diffraction, FTIR and UV-Vis spectroscopic analysis. Single-crystal X-ray diffraction revealed that cluster **1** presents a supramolecular interlocking chain structure. Similar cluster 4-$PhPy_4Cu_4I_4$ (**2**) was also prepared and re-determined. The TD-DFT calculations reveal that their UV-Vis absorption and luminescence originate from the hybrid [(X + M)LCT] excited states. Based on the cluster **1**, a paper-based sensor (**1**) was prepared through a composite process, it shows a remarkable PL quenching response for pyridine (Py) / cyclohexylamine (CYA) detection with good selectivity. Using this sensor, an amazing sensing speed of T_{90} = 5 s was achieved for the detection of Py and T_{90} = 10 s for CYA. In the fluorescence sensing response of these two volatile organic compounds, their maximum fluorescence quenching efficiency reached outstanding 98%. The sensor (**2**) based on cluster **2** also demonstrated similar fluorescence sensing performance for selective detection of Py or CYA.

Keywords Cubane-type Cu_4I_4 cluster · Crystal structure · Fluorescence sensing · VOCs · TD-DFT calculation

Introduction

Copper (I) is a readily available and applied species in coordination chemistry, easily forming complexes or clusters, and because of the relatively small energy differences between the various possible isomers, it exhibits an extremely rich structural diversity, including coordination number and stoichiometric changes [1–3]. Halogen ions can also have multiple coordination modes such as terminal and μ_2-, μ_3-, μ_4-bridges [3–7], so the combination of halogen with Cu(I) and peripheral organic ligands leads to a more intriguing CuI cluster-based material chemistry, with the general formula of $Cu_xI_y(L)_z$ (L = N, S, or P based organic ligand). In fact, in addition to the interesting structural diversity of CuI clusters, more attention is drawn to their rich luminescence properties and their attractive numerous innovative applications. For example, Liu G.-N. et al. previously reported a case of stair-step Cu_4I_4 cluster with good fluorescence sensing performance for nitrobenzene and other compounds [8]. Recently, Mohammed O.F. et al. developed a series of highly luminescent inks and hybrid scintillators based on CuI cluster and successfully implemented high-resolution X-ray imaging detection [9, 10].

Of course, among the many types of CuI cluster structures, rhomboid Cu_2I_2 cluster and cubane-like Cu_4I_4 cluster are the two that have been investigated extensively [3, 4, 9]. It is because the former has been found to generally have a high luminescent efficiency, which is due to the rigidity of the Cu_2I_2 cluster core structure and the mechanism of [(X + M)LCT] excited state luminescence. The latter material, owing to its more abundant luminescent excited states, exhibits not only intrinsic luminescence but also demonstrates a highly intriguing luminescent response behavior to diverse external stimuli. Kirakci K. et al. report intensive visible light radioluminescence upon X-ray irradiation of cubane-like Cu_4I_4 clusters containing triphenylphosphine

✉ Li Song
songli@zstu.edu.cn

Wen-Xiang Chai
wxchai@cjlu.edu.cn

1 College of Materials and Chemistry, China Jiliang University, Hangzhou 310018, P.R. China

2 School of Chemistry and Chemical Engineering, Zhejiang Sci-Tech University, Hangzhou 310018, P.R. China

Published online: 05 April 2025

or pyridine ligands in the solid state [11]. In fact, since the 1970s, a series of studies have reported a variety of interesting luminescence phenomena of $Cu_4I_4Py_4$, including the thermochromism of luminescence [12]. Later, De Angelis, F and Ford P.C. et al. studied the structural mechanism of its luminescence through TD-DFT calculation, and further clarified the metal-to-ligand charge transfer (MLCT) and cluster-centered (CC) excited state luminescence [13]. Subsequently, through ligand design, a variety of sol-gel and thin film devices based on cubane-like Cu_4I_4 clusters were prepared. It was again found that the luminescence properties of thermochromic and mechanochromic were affected by Cu-Cu interactions [14, 15]. In addition, interesting phenomena based on Cu_4I_4 clusters also include luminescence vapochromism and medium rigidity effects, which fully indicate the potential of these materials for intelligent environmental sensing applications [3]. The organic ligands that allow convenient design and selection can provide sufficient room for regulation. Therefore, the Cu_4I_4 cluster is a good structural platform for the development of new intelligent fluorescent responsive materials.

Previously, we successfully developed a paper-based composite film sensor for ammonia detection by using rare earth complexes as fluorescent probes [16]. Then, two simple $PCuI_2CuP_2$-type clusters (P is tri(p-tolyl)phosphine / tri(m-tolyl)phosphine) were used to prepare sensors that exhibited vapochromism and enhanced fluorescence response to pyridine vapor [17]. Of course, other kinds of Cu(I) complexes are also being tried to develop sensors for the detection of VOCs (volatile organic compounds) such as pyridine, cyclohexylamine and acetophenone [18–20]. Encouraged by these works, we have made some new explorations on VOCs sensing based on CuI clusters. Herein, we report the synthesis, crystal structure and TD-DFT calculation of two cubane-like Cu_4I_4 clusters, 3-$PhPy_4Cu_4I_4$ (**1**) and 4-$PhPy_4Cu_4I_4$ (**2**) (Scheme 1). At the same time, based on the paper-based film process [16], the application of these two clusters in VOCs fluorescence sensing has also achieved the expected positive results.

Experimental

Materials and General Methods

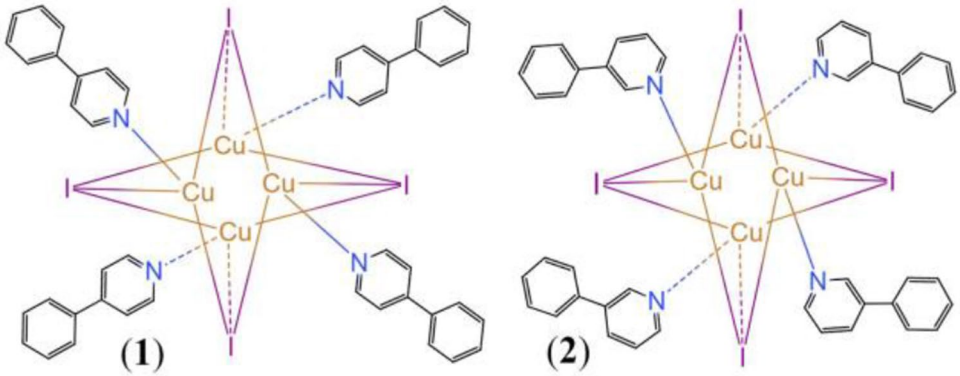
All reagents were purchased commercially and used without further purification. Elemental analyses for C, H and N were performed on an Elementar Vario MICRO analytic instrument. Infrared spectra were recorded on a Bruker Tensor27 FT-IR spectrophotometer in the range of 4000–400 cm^{-1} in KBr lamella. The UV-Vis diffuse reflectance spectra (using $BaSO_4$ as reference substance) were recorded on the as-synthesized powder samples with a Shimadzu UV3600 spectrometer at room temperature. The fluorescence spectra were measured on a NOVA high sensitive spectrometer (ideaoptics, China).

Synthesis of 3-$PhPy_4Cu_4I_4$ (1)

With continuous stirring, 5 mL ethanol solution containing 0.155 g (1 mmol) of 3-phenylpyridine (3-PhPy) was added into the CH_3CN (15 mL) suspension containing 0.190 g CuI (1 mmol). The reaction liquid was heated to 82 °C under continuous stirring and then refluxed for 3 h, whereupon the reaction liquid color was light yellow. It was then cooled to room temperature, where a large amount of precipitation formed. The product was filtered and dried. Yield: 52.6%. IR (KBr, cm^{-1}): 3051 (w), 3028 (w), 2972 (w), 2918 (w), 2888 (w), 2849 (w), 1609 (s), 1538 (w), 1486 (ms), 1415 (s), 1221 (m), 1154 (w), 1073 (m), 1010 (w), 918 (w), 837 (m), 763 (s), 729 (m), 695 (m), 617 (m), 559 (w). Anal. Calc. for $C_{44}H_{36}Cu_4I_4N_4$: C, 38.22; H, 2.62; N, 4.05. Found: C, 38.49; H, 2.42; N, 4.13%. The single crystals used for X-ray diffraction study are obtained by volatilization and recrystallization in dichloromethane.

Synthesis of and 4-$PhPy_4Cu_4I_4$ (2)

Previously, Tong etc. have reported a room temperature solution reaction synthesis method of the complex (reported yield of 14%) [21]. CuI cluster **2** can also be prepared



Scheme 1 Schematic structures of 3-$PhPy_4Cu_4I_4$ (**1**) and 4-$PhPy_4Cu_4I_4$ (**2**)

following a similar procedure to that for **1**, only using 4-phenylpyridine (4-PhPy) instead of 3-PhPy. Yield: 83.4%. The single crystals used for X-ray diffraction study are obtained by volatilization recrystallization. A large amount of powder products can be obtained by rapid cooling crystallization method, and the yield of **2** is significantly increased compared with the methods in the literature [21]. The large increase in yield should be due to the fact that the reaction solvent used is significantly less soluble at room temperature than dichloromethane.

Preparation of paper-based Sensor Containing Complexes

A paper-based VOCs sensor was fabricated on commercial filter paper without further treatment. A drop-casting method is used to prepare paper-based sensors using optimal probe solution as raw material [16–19]. The optimal probe solution was made by dissolving the CuI clusters **1** / **2** (10 wt%) and PMMA in dichloromethane. After being dried under reduced pressure, the paper-based sensor was cut in to disc-shaped sensing sheet (Φ = 5 mm). In order to achieve fluorescent sensing detection of VOCs, the sensing sheet is placed on the top of the Y-type (6 + 1) optical fiber probe.

X-ray Crystallography

Suitable single crystals of two clusters were selected for X-ray diffraction analyses. Diffraction intensities for these clusters were collected at 170(2) K on a Bruker APEX-II CCD (Mo Kα, λ = 0.71070 Å for **1**) and D8 Venture (Ga Kα, λ = 1.34139 Å for **2**) diffractometer with the ω-scan technique (see Table S1). The structures were solved with the direct method and refined with full-matrix least-squares using the SHELX package [22, 23] and OLEX2 software [24]. Anisotropic displacement parameters were applied to all non-hydrogen atoms, and the organic hydrogen atoms were generated geometrically (C–H 0.96 Å) and refined using a riding model. The selected bond length and angle parameters are collected in Table S2 and S3.

Theoretical Calculations

The calculations described here were performed with the Gaussian 09 (G09) program package [25]. Both geometries of singlet ground state (S_0) for CuI clusters **1**–**2** were fully optimized by means of density functional theory (DFT) calculations. In order to gain detailed insight into the electronic origin of the absorption and emission properties, time-dependent density functional theory (TD-DFT) calculation was performed on the calculated S_0 geometry. The three-parameters hybrid B3LYP exchange-correlation functional and a settled combination of basis sets were used throughout [13, 26]. The basis sets combination includes the relativistic effective core potential standard basis set LANL2DZ for Cu and I, and the standard 6-31G* basis set for C, H and N. In addition, polarization function was also added for I [ξ(d) = 0.266] [2, 27]. The TD-DFT calculation can estimate the vertical electron excitation energies including some account of electron correlation. To get more insight into the absorption/emission transition, wavefunction analysis is done between excited states and ground state by using the Multiwfn program [28, 29]. The excitation energies, oscillator strength parameters and corresponding major orbital contributions for some main electronic transitions are gathered in Table S4 for two clusters.

Results and Discussion

Description of Crystal Structure

Single-crystal X-ray diffraction (SCXRD) shows that the cluster **1** ($3\text{-PhPy}_4Cu_4I_4$) is isolated tetranuclear CuI cluster with Cu_4I_4 core. As shown in Fig. 1a, the asymmetric crystallographic unit is a binuclear cluster, which is exactly half of the entire tetranuclear cluster ($3\text{-PhPy}_2Cu_2I_2$). In the asymmetric unit, the Cu_2I_2 cluster core presents a butterfly structure centered on the Cu—Cu connection, where the dihedral angles of the two Cu2I planes are 144.41(3)°. Then, two such asymmetric units are aggregated in a cross-stacking fashion to form a tetranuclear cluster structure (Fig. 1b). In this Cu_4I_4 cluster, both types of copper atoms adopt CuI_3N tetrahedral coordination modes. Previously, Angelis and Ford et al. revealed through TDDFT calculations that copper-copper interactions play an important role in the luminescence of the $[Cu_4I_4(\text{pyridine})_4]$ cluster [13]. In this cluster **1**, every two coppers are also connected with a distinct copper-copper interaction to form a unified tetrahedral Cu_4 core, since the copper-copper distances (Table S2) are smaller than the sum of their van der Waals radii (2.80 Å) [30]. According to the optimized geometric structure, the Wiberg bond order value of these interaction are about 0.2 (see Table S3), which also indicates the existence of copper-copper interaction [31]. In this cluster, the lengths of Cu–I are between 2.6467(7) Å and 2.7950(7) Å, and two Cu–N bonds are 2.024(4) Å and 2.031(4) Å, respectively. As shown in Table S2, these bond lengths are similar to those of previously reported CuI clusters [6, 17, 32–34]. In crystal structure 1, there are no classic hydrogen bond between clusters but an interesting interlocking supramolecular structure between adjacent clusters appears in the *b*-axis direction. (Fig. 1c). Shown in the illustrated space-filling mode, the supramolecular cable-chain like structure

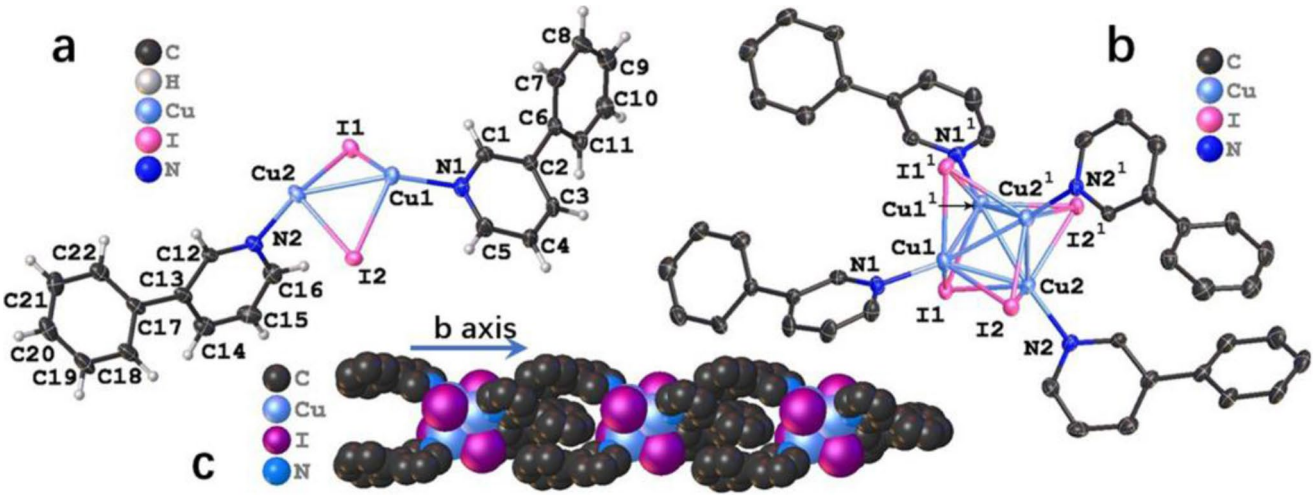


Fig. 1 The crystal structure of cluster **1**: (**a**) the crystallographic independent unit with thermal ellipsoids at 50% probability, (**b**) the tetranuclear cluster structure, (**c**) the supramolecular cable-chain like structure along *b* axis

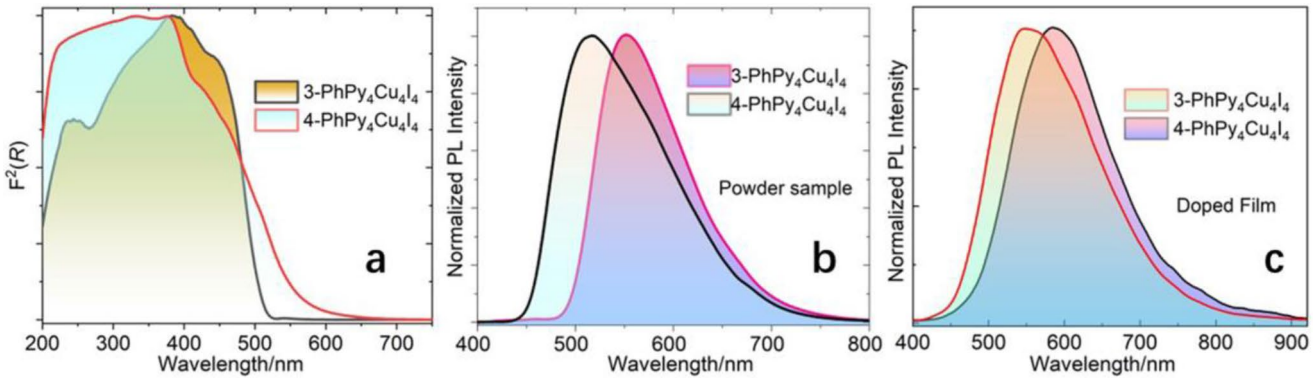


Fig. 2 The UV absorption spectra of clusters 3-$PhPy_4Cu_4I_4$ (**1**) and 4-$PhPy_4Cu_4I_4$ (**2**) (**a**); and the PL emission spectra (λ_{ex} = 365 nm) of clusters **1** and **2** recorded on powder samples (**b**) and doped film samples (**c**)

can be clearly seen. The packing diagram of structure 1 is shown in Figure S1. Although the crystal structure of cluster **2** has also been re-determined at 170 K, given the similarity of the reported data (CCDC XALTOU) [21], it will not be described here.

UV-Vis Absorption and PL Spectra

The UV-Vis absorption spectra of cluster **1**–**2** recorded on as-synthesized powder samples are shown in Fig. 3. Prior to this test, the phase of the powder sample was confirmed by powder X-ray diffraction (PXRD) experiments (Figure S2 and S3). In the spectra of **1** and **2**, only one broad absorption band with some shoulder is found in the range of 200 nm to about 520 nm (for **1**) or 650 nm (for **2**). According to the previous works [6, 13, 17, 34], this broad absorption band should be assigned to the π–π* transition involving the phosphine ligand, as well as components from the metal to ligand charge transfer (MLCT) and / or halide to ligand charge transfer (XLCT), or cluster-centered (CC) transition. In the case of cluster **1**, the acromion absorption (about 250 nm) in the high energy region should be assigned to the π–π* transition involving the 3-PhPy ligand. The maximum absorption (about 400 nm) and its other acromion absorption (about 450 nm) band should be assigned to the MLCT and / or XLCT or cluster-centered transition. In addition, it should be pointed out that the absorption edge of cluster **2** is significantly redshifted than that of cluster **1**.

Under the excitation of ultraviolet light (Figure S4 and S5), as shown in Fig. 3b, both powder samples of two clusters show good fluorescence properties. Their fluorescence emission showed a single broad peak, and no obvious vibrational fine structure was observed. However, comparing the two clusters, it is found that the order of the wavelength of the emission peak and the absorption edge is inconsistent. The fluorescence spectra of PMMA composite films doped with cluster **1** / **2** were also measured (Fig. 2c). The luminescence of doped films has a certain degree of red shift relative to the crystalline samples. This common phenomenon should be attributed to the fact that the excited state of the

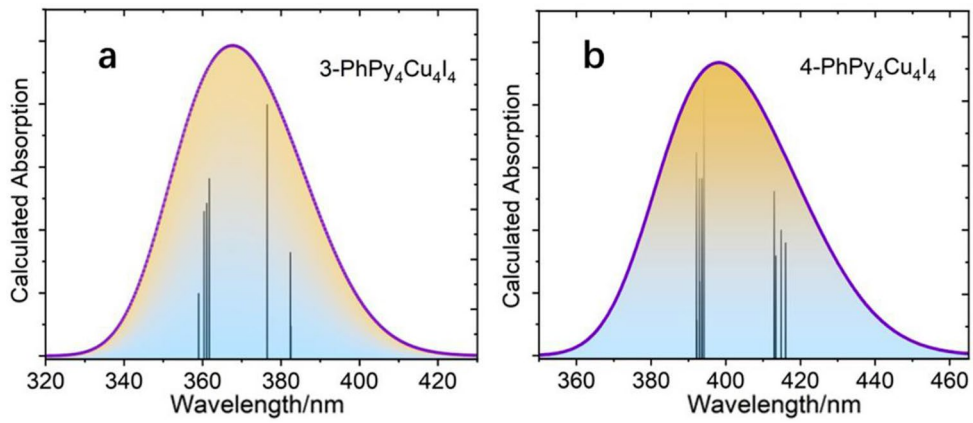


Fig. 3 The calculated electronic absorption spectra of clusters (**a**) 3-$PhPy_4Cu_4I_4$ (**1**) and (**b**) 4-$PhPy_4Cu_4I_4$ (**2**) focused on the ten lowest energy transitions

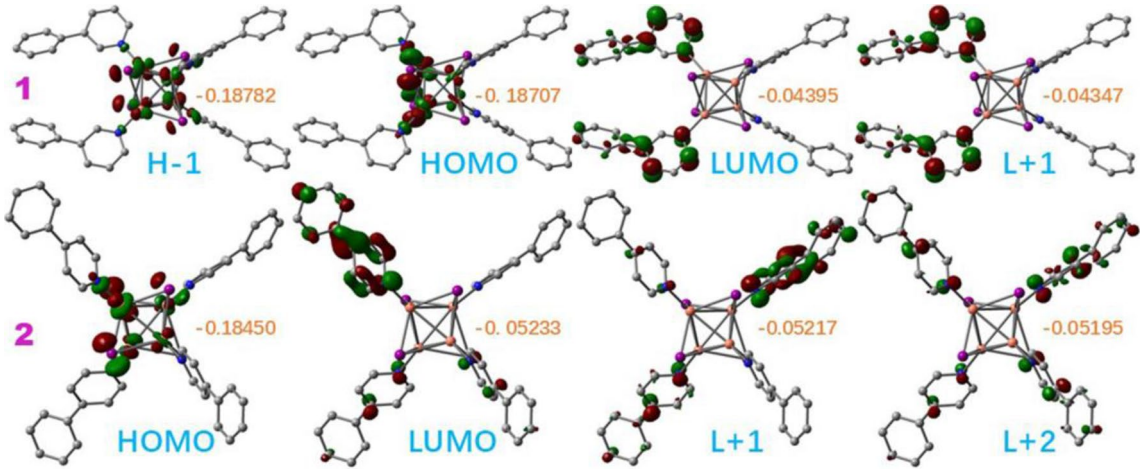


Fig. 4 Some frontier orbitals plots of clusters **1**–**2** based on TD-DFT calculations, isovalue = 0.03, H = HOMO and L = LUMO

doped cluster molecule has a greater degree of vibrational freedom in the film [35]. However, the blue shift of the emission peak of the crystalline powder of cluster **2** is particularly significant relative to the doped state. This particular blue shift even causes the wavelength value of the emission peak to change from **2** > **1** in the doped state (the same as the absorption edge) to **1** > **2** in the powder state. Based on the comprehensive analysis of various current luminescence mechanisms, this special blue shift should be related to the aggregation induced emission (AIE) mechanism [36, 37]. Because the relaxation of excited molecules is limited in the aggregation state of the crystal, especially the rotation of the aromatic ring in the ligand is greatly limited.

TD-DFT Calculation

In order to get a better insight into absorption and luminescence spectra of cluster **1**–**2**, the simulated absorption spectra and their related excited states and orbital compositions have been calculated at B3LYP level by density functional method. Using TD-DFT calculations based on the optimized geometries, the positions of the first 10 electronic transitions are obtained and used to generate bar graphs and calculate spectra by applying a thickness of 0.1 eV to each bar graph (Fig. 3). Table S4 lists the positions, oscillator strengths, and excited state composition of these 10 singlet transitions, as well as some low-energy triplet transitions. For luminescent materials, the first singlet / triplet excited state (S_1 / T_1) and its transition composition are very important to discuss both photochemical and photophysical properties. [2, 27, 38]. As shown in Table S4, the S_1 transitions of both clusters are composed of multiple transition modes, *i.e.*, **1** consists of transition modes HOMO→L+1 (61.3%) and H-1→LUMO (25.9%), while **2** consists of transition modes HOMO→LUMO (58.5%), HOMO→L+2 (19.1%), HOMO→L+1 (9.5%). HOMO (H) and LUMO (L) are namely the highest occupied molecular orbital and the lowest unoccupied molecular orbital. As shown in Fig. 4, the composition of the frontier orbitals of these two clusters are very similar. Those occupied frontier orbitals of the two clusters, both H-1 and HOMO, are made up of Cu-d orbitals and I-p orbitals of their cluster cores. In contrast, those unoccupied orbitals, LUMO & L+1 for **1**, and LUMO, L+1 & L+2 for **2**, all consist of the phenyl-π* and pyridine-π*

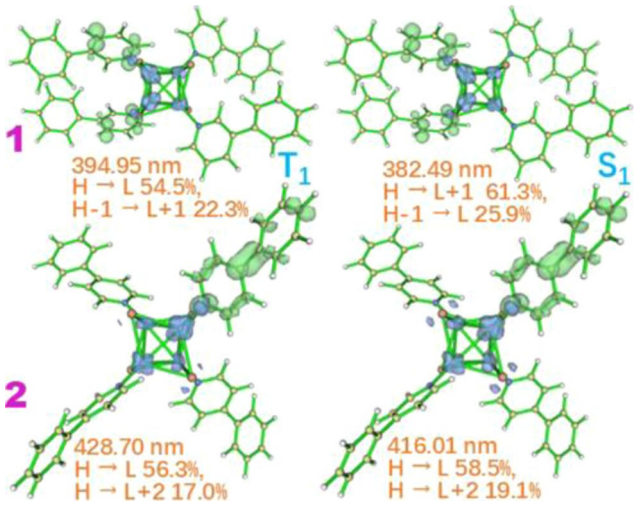


Fig. 5 Electron density difference plots of T_1 and S_1 transition in **1**–**2**. (Electron densities move from the blue area to the green area; isovalue = 0.004)

orbitals of two PhPy ligands. Therefore, these S_1 transitions feature significant charge transfer characteristics from the Cu-d and I-p orbitals to the π* orbitals of peripheral ligand. In order to more intuitively understand the charge separation characteristics of the S_1 state, we use the electron difference diagram between the ground state and the S_1 excited state to show the spatial separation of electrons and holes. As shown in Fig. 5, the S_1 states clearly shows the charge transfer from the cluster core to the peripheral PhPy ligand. Therefore, the S_1 states should both be assigned as a mixed $[(X+M)LCT]^1$ excited state. Moreover, the charge separation in T_1 is almost the same as in S_1, which is called the $[(X+M)LCT]^3$ excited state. It can also be seen from the figures that the electrons in **2** are much more delocalized than the electrons in **1**. Additionally, some other main low-energy transitions show same charge transfer characteristic (Figure S6 and S7), as in the S_1/T_1 transitions described above. It is obvious that the luminescence and low-energy electronic absorption of these two clusters originate from the [(X + M)LCT] excited states.

PL Sensing Performance for VOCs

In order to explore the sensing properties of clusters **1**–**2** to VOCs, we have developed a paper-based film sensor based on these clusters using the previous method [16]. Due to the volatility and often biological toxicity of VOCs, their detection is of great significance. For instance, pyridine (Py) and cyclohexylamine (CYA) are two typical VOCs. Py is widely used as a raw material in the production of vitamins, pharmaceuticals, insecticides, and plastics, and is also widely used in industrial production as a solvent and catalyst. CHA is a common indoor air contaminant because it is widely used as a corrosion inhibitor, such as in urban heating boilers and pipeline systems. When the paper-based sensor uploaded with cluster **1** is exposed to pyridine (Py), its PL emission exhibits a rapid and obvious quenching process, although the peak wavelength is unchanged (Fig. 6a). After exposure to Py for only 5 s, the PL intensity of the paper-based sensor (**1**) decreases by more than 90% (Figure S8). When the exposure time is extended to 5 min, the value will continue to rise to about 98% and reach a level close to saturation. That is to say, for Py sensing, the T_{90} value (T_{90} refers to the time required for the gas detector to change from reading 0 to 90%) [39] can be as low as about 5 s. Therefore, this sensor (**1**) can be used as a convenient and rapid response tool for the pyridine vapor detection. In addition, the sensor also exhibits fluorescence sensing performance of a quenched response type for cyclohexylamine (CYA) vapor. As shown in Fig. 6b, the PL emission of the sensor (**1**) presents a rapid and obvious quenching process under CYA vapor expose. After exposure to Py for only 5 s, the PL quenching of the sensor (**1**) rapidly reaches a very high value of 80% (Figure S9). When the response time is extended to 5 min, the quenching efficiency value will continue to rise to about 98%, reaching a level near saturation.

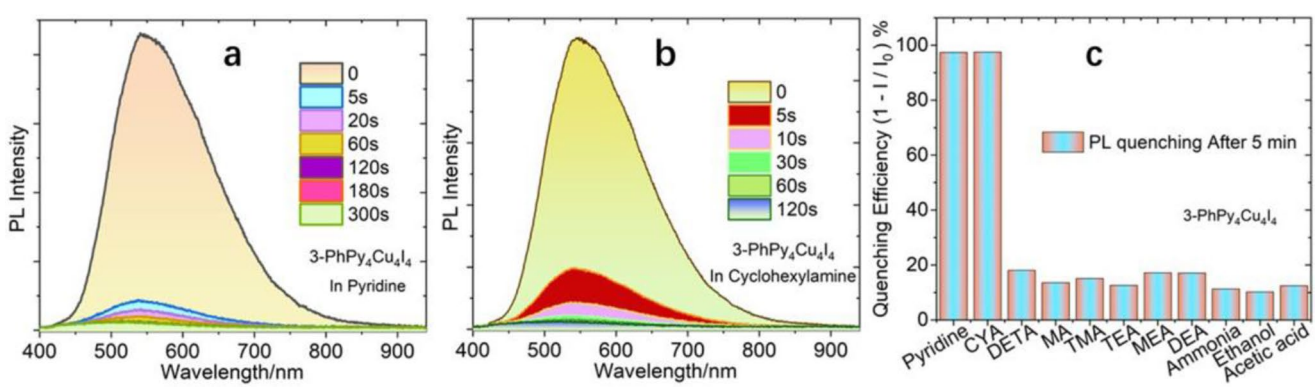


Fig. 6 Time resolved PL spectra of the sensor (**1**) before and after exposure to (**a**) Py and (**b**) CYA for a period of time (λ^{ex} = 365 nm, λ^{em}_{max} = 540 nm), (**c**) Selective detection of Py and CYA based on quenching effect

Therefore, the fluorescence response of the sensor (**1**) to CYA is also very fast, with a T_{90} value [39] as low as 10 s.

In order to evaluate the selectivity of the sensor (**1**) in Py / CYA detection, a series of tests were performed to study the interference of different common vapors on PL quenching effect ($1-I/I_0$). As a result, common vapor molecules, whether oxygen-containing or nitrogen-containing coordination molecules, or alkaline, acidic or neutral gas molecules, do not show significant PL quenching effect. As shown in Fig. 6c, even the vapor of different amines such as DETA (diethylenetriamine), MA (methylamine), TMA (trimethylamine), TEA (triethylamine), MEA (ethanolamine), DEA (diethanolamine), ammonia, ethanol, and acetic acid has little interference with the detection. And some experiments have found that the PL quenching by Py / CYA was about 98%, but there is no significant change after other common vapors were respectively introduced into the detection system. On the contrary, when the sensor's luminescence is stabilized in TEA vapor, and then placed in Py / CYA vapor environment, it is found that the fluorescence quenching efficiency rises rapidly to nearly 100%. These results indicated that this sensor loaded with cluster **1** had good anti-interference ability and selectivity in the application of Py / CYA vapor detection.

Different from sensor (**1**), when the paper-based sensor loaded with cluster **2** is exposed to Py / CYA vapor, its PL emission shows both a quenching process and a vapochromic process (blue shift of luminescence). As shown in Fig. 7a and b, when the sensor is placed in Py or CYA vapor environment, its luminescence also shows a typical quenching response, and can reach about 95% quenching efficiency after 5 min. In contrast, the rate of sensing response of sensor (**2**) is much slower than that of sensor (**1**). This may be related to the difference of steric hindrance of probe molecules in the two sensors, or it may be caused by different sensing response mechanisms. In addition, the selective detection of the sensor (**2**) is also evaluated (DIPA = diisopropylamine). As shown in Fig. 7c, the sensor (**2**) also exhibits good anti-interference ability and selectivity in the detection of Py / CYA vapor.

Proposed Mechanism of PL Sensing

Fluorescence quenching is defined as the decrease in the fluorescence intensity of a fluorescent material due to processes like the internal filter effect (IFE), intramolecular charge transfer (ICT), photoinduced electron transfer (PET), fluorescence resonance energy transfer (FRET), exciplex formation, or a chemical reaction between the fluorophore and the quencher [16, 40–44]. The UV absorption of pyridine is below 270 nm, while the excitation and emission spectra of cluster **1** do not overlap with this, as its peak wavelengths are 400 nm and 540 nm, respectively. Obviously, neither IFE nor FRET mechanisms are suitable for explaining the sensing mechanism of cluster **1** [19, 41]. In view of the rapid sensing response of cluster **1** to pyridine, and the solid-state fluorescence of cluster $Cu_4I_4Py_4$ is not weak [13], the static response mechanism for the ligand substitution reaction is also inappropriate. The response type formed by excimer complexes will show a ratio fluorescence sensing pattern, the bathochromic shift was caused by the formation of exciplexes. Therefore, the most reasonable response pattern should be the PET mechanism. According to the DFT calculation on a B3LYP/6-31G* level, the HOMO and LUMO orbital energies of pyridine are −0.25262 a.u. and −0.02242 a.u. respectively. As demonstrated in Scheme 2a, the LUMO energy level of pyridine is very close to that of **1** (-0.04395 a.u.). Therefore, the electrons can be transferred from the LUMO of the excited **1** to the LUMO of pyridine, indicating the presence of PET. Through this pathway, the excited molecules are finally deactivated without radiation, leading to fluorescence quenching (a conceptual diagram shown in Scheme 2). Of course, since the LUMO level of cluster **1** is slightly lower than that of pyridine, it is more likely that pyridine first approaches the 3-PhPy ligand on the cluster to form the exciplex, and the LUMO orbitals of **1** and pyridine

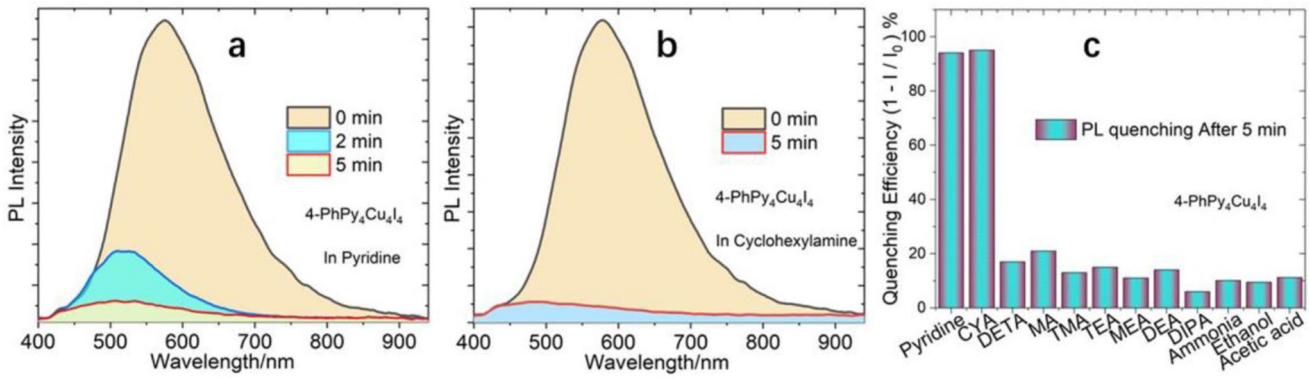
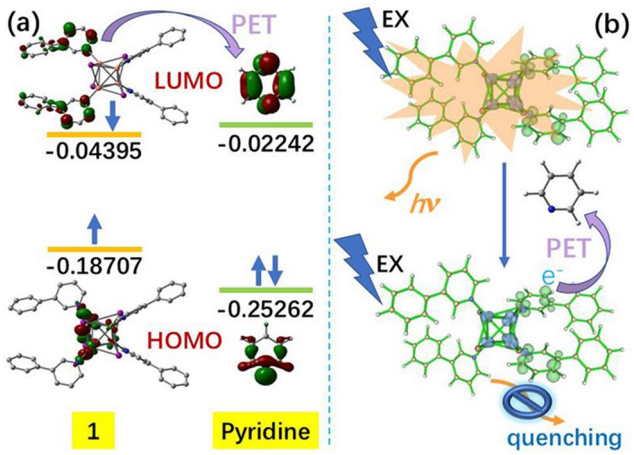


Fig. 7 Time resolved PL spectra of the sensor (**2**) before and after exposure to (**a**) Py and (**b**) CYA (λ^{ex} = 365 nm, λ^{em}_{max} = 580 nm), (**c**) Selective detection of Py and CYA based on quenching effect



Scheme 2 (**a**) The PET process and calculated frontier orbital energy levels of cluster **1** and pyridine, and the orbital maps of their HOMOs & LUMOs; and (**b**) Proposed mechanism of PL quenching of **1** by pyridine. The molecular orbital energies are in atomic units, and the isovalue is 0.05

in the exciplex should be flipped to enable a fast PET process [44]. Considering the similarity of molecular structure and properties, the fluorescence quenching mechanism of **2** should also be similar.

Conclusion

In summary, we have successfully designed and synthesized a cubane-type Cu_4I_4 cluster 3-$PhPy_4Cu_4I_4$ (**1**) with 3-PhPy as peripheral ligand, which shows rapid fluorescence sensing for Py and CYA. The supramolecular interlocking chain structure of cluster **1** was revealed by SCXRD, and the low temperature structure of cluster 4-$PhPy_4Cu_4I_4$ (**2**) was also obtained. The TD-DFT calculations reveal that their UV-Vis absorption and luminescence originates from the hybrid [(X+M)LCT] excited states. Based on cluster **1**, we successfully prepared paper-based sensor through the composite process, and found that sensor (**1**) shows a remarkable PL quenching response for Py / CYA detection with good selectivity. The fluorescence quenching efficiency for the Py sensing response can reach 98%, and the T_{90} value shows an amazing speed of 5 s. For CYA sensing, these values can also reach 98% and 10 s, respectively. The sensor (**2**) based on cluster **2** also demonstrated similar fluorescence sensing performance for selective detection of Py or CYA.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10876-025-02811-4.

Acknowledgements The authors are grateful for financial support from the Project of Innovation and Entrepreneurship Training for National Undergraduates (No. 202410356057), Fundamental Research Funds for the Provincial Universities of Zhejiang (No. 2023YW104).

Author Contributions Xin-Yang Du: Writing-Original draft preparation, Formal analysis, Investigation Li Song: Writing-Reviewing and Editing, Methodology, Visualization, Conceptualization, Validation Xin-Yu Wei: Visualization, Formal analysis, Investigation, Data curation Ying-Ying Zhang: Visualization, Data curation Yu-Xin Zhang: Visualization, Formal analysis Xia-Qun Zeng: Formal analysis, Methodology Hao-Nan Hu: Visualization Zhou-Hang Jiang: Investigation Yong Zhang: Investigation Wen-Xiang Chai: Supervision, Conceptualization, Methodology, Writing-Reviewing and Editing, Project administration, Funding acquisition.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

1. M. G. Crestani, G. F. Manbeck, W. W. Brennessel et al (2011) Inorganic Chemistry 50, 7172–7188
2. W. Chai, M. Hong, L. Song et al (2015) Inorganic Chemistry 54, 4200–4207
3. J. Troyano, F. Zamora, S. Delgado (2021) Chem Soc Rev 50, 4606–4628
4. J. Conesa-Egea, F. Zamora, P. Amo-Ochoa (2019) Coordination Chemistry Reviews 381, 65–78
5. L. Maini, D. Braga, P. P. Mazzeo et al (2012) Dalton Transactions 41, 531–539
6. R.-E. Shou, W.-X. Chai, L. Song et al (2017) J Clust Sci 28, 2185–2203
7. Z.-X. Liu, J.-Q. Wang, Y. Mu et al (2020) Inorganic Chemistry 59, 17870–17874
8. G.-N. Liu, R.-D. Xu, R.-Y. Zhao et al (2019) ACS Sustainable Chemistry & Engineering 7, 18863–18873
9. Y. Zhou, T. He, P. Yuan et al (2023) ACS Materials Letters 5, 2002–2008
10. T. He, Y. Zhou, P. Yuan et al (2023) ACS Energy Letters 8, 1362–1370
11. K. Kirakci, K. Fejfarová, J. Martinčík et al (2017) Inorganic Chemistry 56, 4609–4614
12. P. C. Ford, E. Cariati, J. Bourassa (1999) Chemical Reviews 99, 3625–3647
13. F. De Angelis, S. Fantacci, A. Sgamellotti et al (2006) Inorganic Chemistry 45, 10576–10584
14. C. Tard, S. Perruchas, S. Maron et al (2008) Chemistry of Materials 20, 7010–7016
15. S. Perruchas, X. F. L. Goff, S. Maron et al (2010) Journal of the American Chemical Society 132, 10967–10969
16. Y.-Y. Wang, L. Song, J.-T. Wang et al (2022) Applied Organometallic Chemistry 36, e6752
17. Z.-Q. Dai, L. Song, Y.-Y. Wang et al (2023) J Clust Sci 34, 2497–2507
18. W.-Z. Xu, L. Song, H.-X. Jin et al (2024) Journal of Molecular Structure 1318, 139283
19. S.-Y. Tang, L. Song, Y.-F. Jia et al (2023) Applied Organometallic Chemistry 37, e7242
20. Y.-F. Jia, L. Song, W.-Z. Xu et al (2023) Journal of Molecular Structure 1292, 136134
21. S. Hu, M.-L. Tong (2005) *Dalton Transactions*, 1165–1167
22. G. Sheldrick (2008) Acta Crystallographica Section A 64, 112–122

23. G. Sheldrick (2015) Acta Crystallographica Section C 71, 3–8
24. O. V. Dolomanov, L. J. Bourhis, R. J. Gildea et al (2009) Journal of Applied Crystallography 42, 339–341
25. G. W. T. M.J. Frisch, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G.A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H.P. Hratchian, A.F. Izmaylov, J. Bloino, G. Zheng, J.L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J.A. Jr. Montgomery, J.E. Peralta, F. Ogliaro, M. Bearpark, J.J. Heyd, E. Brothers, K.N. Kudin, V.N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J.C. Burant, S.S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J.M. Millam, M. Klene, J.E. Knox, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, R.L. Martin, K. Morokuma, V.G. Zakrzewski, G.A. Voth, P. Salvador, J.J. Dannenberg, S. Dapprich, A.D. Daniels, O. Farkas, J.B. Foresman, J.V. Ortiz, J. Cioslowski, D.J. Fox, C.T. Wallingford, Gaussian, Inc., 2009.
26. C. W. Hsu, C. C. Lin, M. W. Chung et al (2011) J Am Chem Soc 133, 12085–12099
27. D.-D. Wang, L. Song, Y.-Y. Wang et al (2020) Applied Organometallic Chemistry 34, e5561
28. T. Lu, F. Chen (2012) Journal of Computational Chemistry 33, 580–592
29. T. Lu, F.-W. Chen (2011) Acta Chimica Sinica 69, 2393–2406
30. E. Cariati, D. Roberto, R. Ugo et al (2005) Inorganic Chemistry 44, 4077–4085
31. O. V. Sizova, L. V. Skripnikov, A. Y. Sokolov (2008) Journal of Molecular Structure: THEOCHEM 870, 1–9
32. P. D. Akrivos, S. K. Hadjikakou, P. Karagiannidis et al (1993) Inorganica Chimica Acta 206, 163–168
33. R. Meijboom (2006) Acta Crystallographica Section E 62, m3511-m3513
34. L. Song, Y.-J. Wang, W.-X. Chai (2019) Inorganic Chemistry Communications 104, 190–196
35. S. Kondo, N. Yoshimura, M. Yoshida et al (2020) Dalton Transactions 49, 16946–16953
36. C. Chen, R.-H. Li, B.-S. Zhu et al (2018) Angewandte Chemie International Edition 57, 7106–7110
37. Y. Wang, H. Gao, J. Yang et al (2021) Advanced Materials 33, 2007811
38. N. Armaroli, G. Accorsi, M. Holler et al (2006) Adv Mater 18, 1313–1316
39. D.-Q. Dai, L. Song, J.-T. Wu et al (2025) Journal of Molecular Structure 1319, 139595
40. D. Genovese, M. Cingolani, E. Rampazzo et al (2021) Chemical Society Reviews 50, 8414–8427
41. E. M. S. Stennett, M. A. Ciuba, M. Levitus (2014) Chemical Society Reviews 43, 1057–1075
42. G. G. Aloisi, F. Masetti, U. Mazzucato (1974) Chemical Physics Letters 29, 502–505
43. Y.-C. Cai, C. Li, Q.-H. Song (2017) ACS Sensors 2, 834–841
44. J. Hirota, K. Usui, Y. Fuchi et al (2019) Chemistry– A European Journal 25, 14943–14952

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.